C Programming

Characters & Strings: Ch.12

char arrays | the NULL terminator | string.h functions

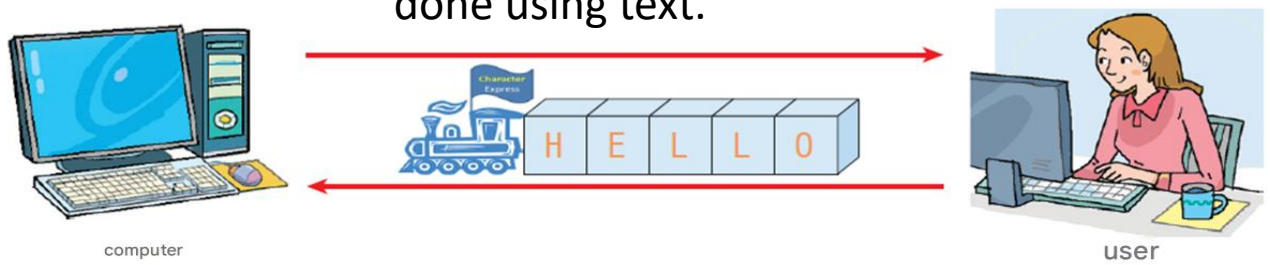
What You Will Learn in This Chapter

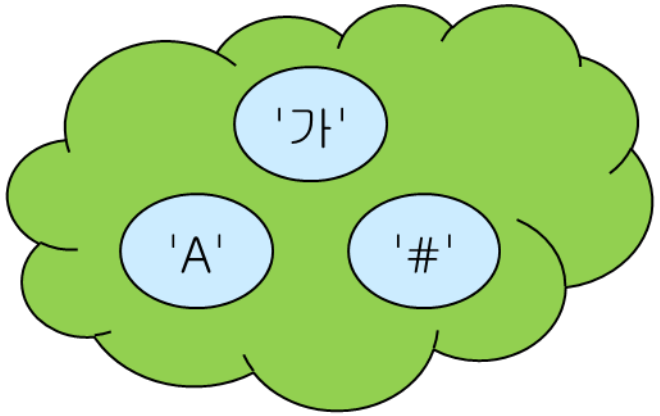
- How to express characters, how to represent strings
- What is a **string**? Input/output of strings
- Character processing library functions
- **String** processing library functions
- Since humans use characters to express information, strings occupy an important position in programs.

How to Represent Strings

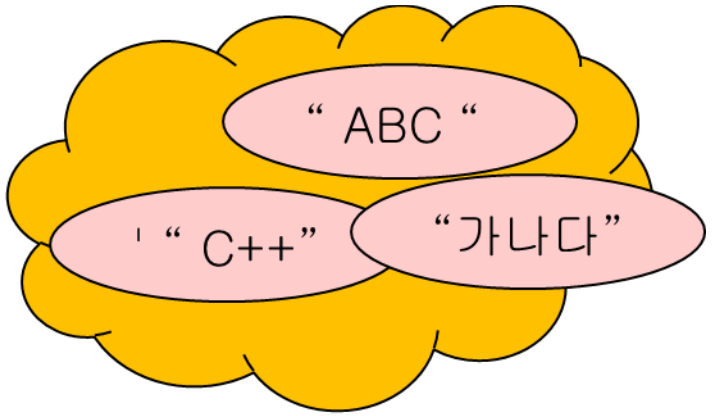
- **String**: a collection of several characters, such as "A", "Hello World!"
- One character is stored in a char type variable.
- Strings are stored as char type arrays: H E L L O

A lot of computer work is done using text.





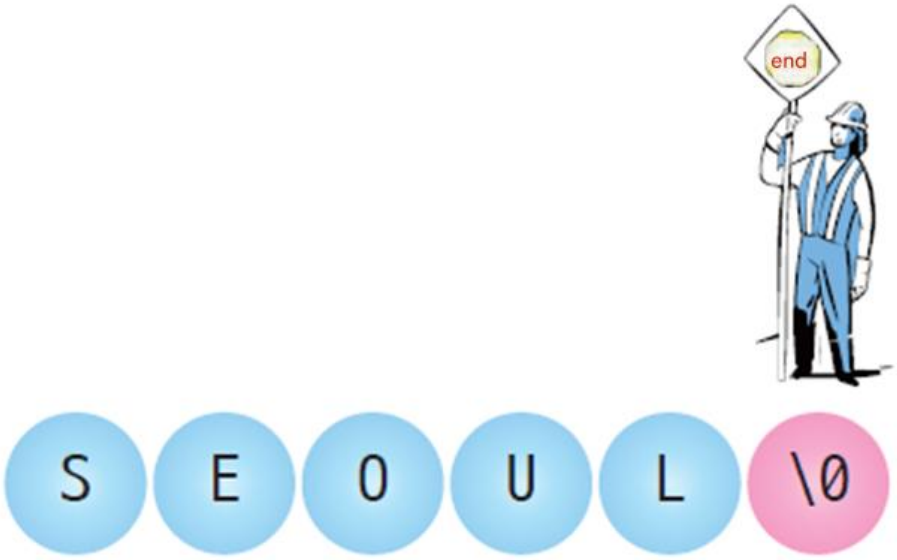
Character

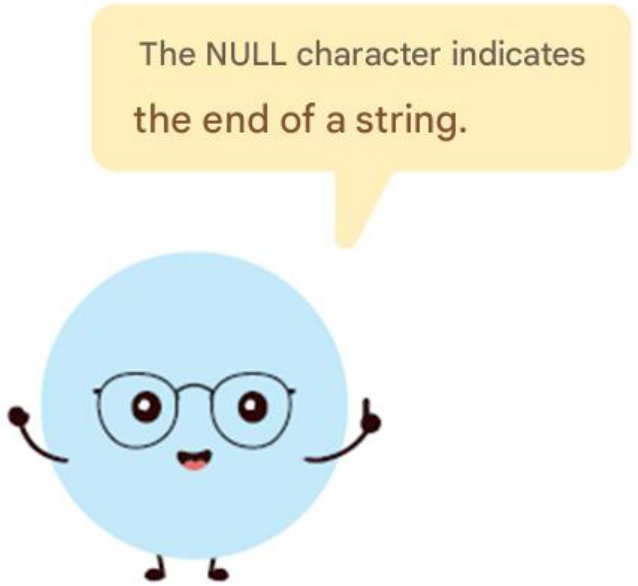


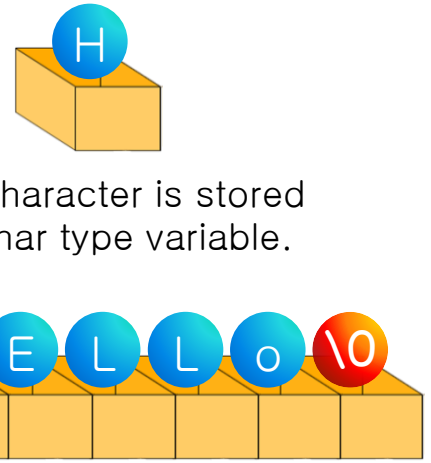
String

How to Represent Strings: NULL Character

- NULL character: indicates the end of a **string**. (They are related but different concepts.)
  '\0' == null character
  NULL == null pointer
  0 == value or null pointer
- Why do we need to mark the end of a **string**?
  Since we don't know where the **string** ends otherwise, we need to mark it.



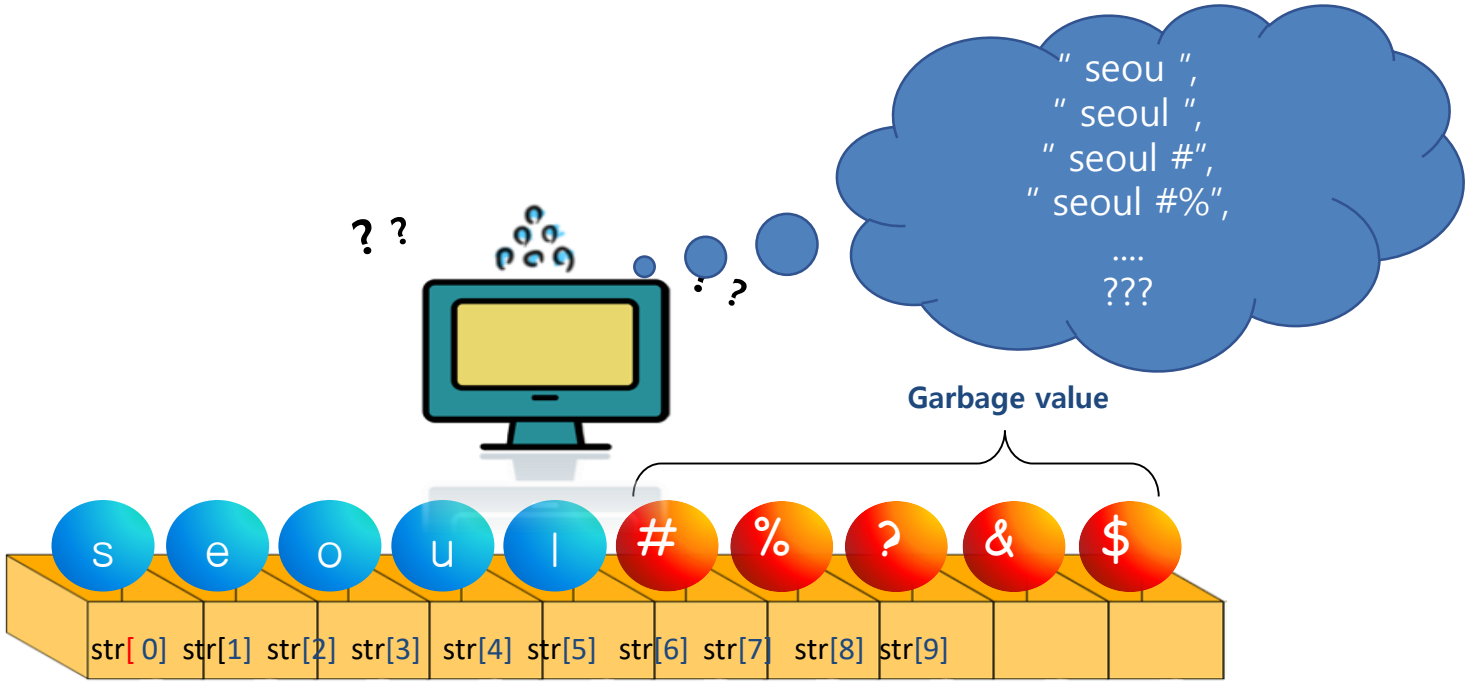




Strings are stored as char type arrays

How to Represent Strings: Why do we need to mark the end of a string?

- Since we don't know where the string ends, we need to mark it.

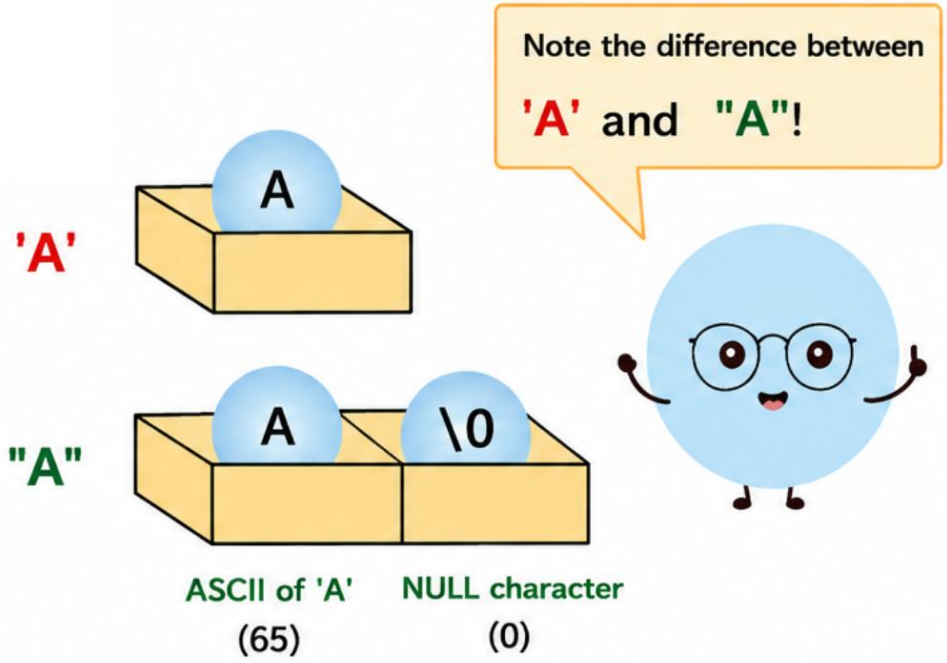


How to Represent Strings: Difference A, ‘A’, “A”

A: The compiler considers A as the name of a variable.

'A': Represents the letter A.

"A": Represents a string consisting only of the letter A. This is different from 'A'.



- The thing to note here is the difference between 'A' and "A".
- 'A' represents a single character and is the same as the ASCII code for the character A.
- "A" is a string, and the NULL character is added to the ASCII code of A to indicate the end of the string.

'A' (single character) : Stored as a single byte → ASCII value of 'A' (65)

"A" (string) : Stored as two bytes → 'A' (65) followed by NULL character '\0' (0)

Example: Printing Characters One by One

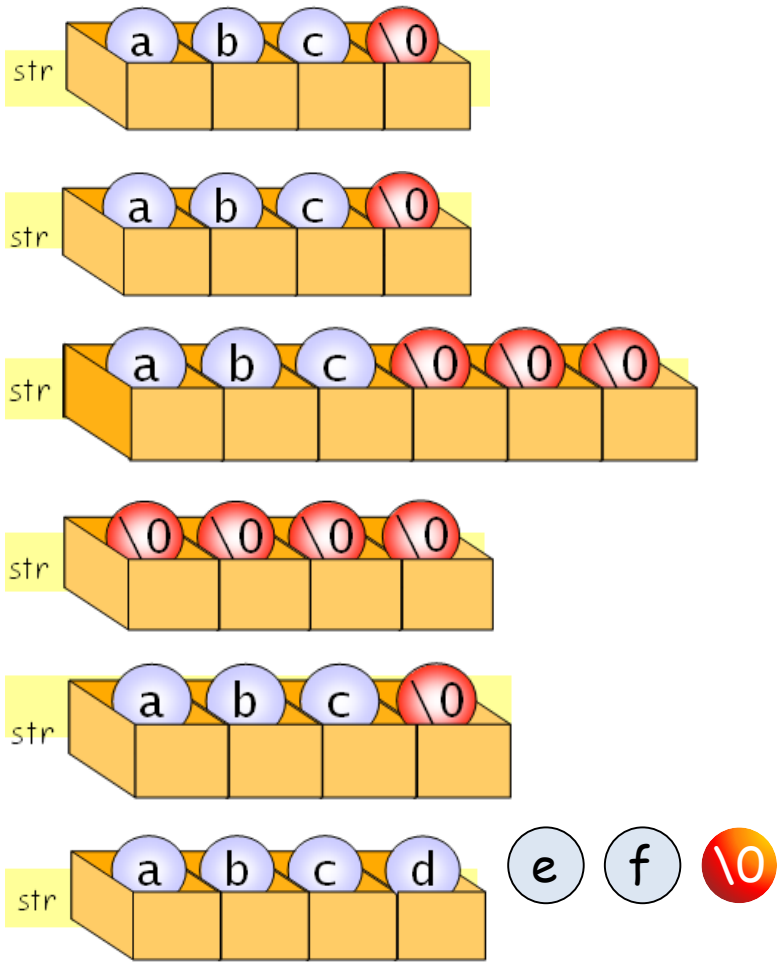
```c
#include < stdio.h >

int main( void )
{
    int i ;
    char str [4];
    str [0] = 'a';
    str [1] = 'b';
    str [2] = 'c';
    str [3] = '\0';

    i = 0;
    while ( str [ i ] != '\0' ) {
        printf ( "%c" , str [ i ]);
        i ++;
    }
    return 0;
}
```

```
abc
```

We print out characters one by one, stopping the loop when a NULL character is encountered.

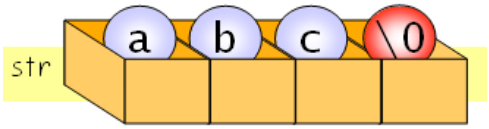
Initializing a character array (String)

- char str [4] = { 'a', 'b', 'c', '\0' };
- char str [4] = "abc";
- char str [6] = "abc";
- char str [4] = "";
- char str [] = "abc";
- char str [4] = "abcdef"; // Error
- char str []; // Error: The compiler doesn't know how much memory to allocate

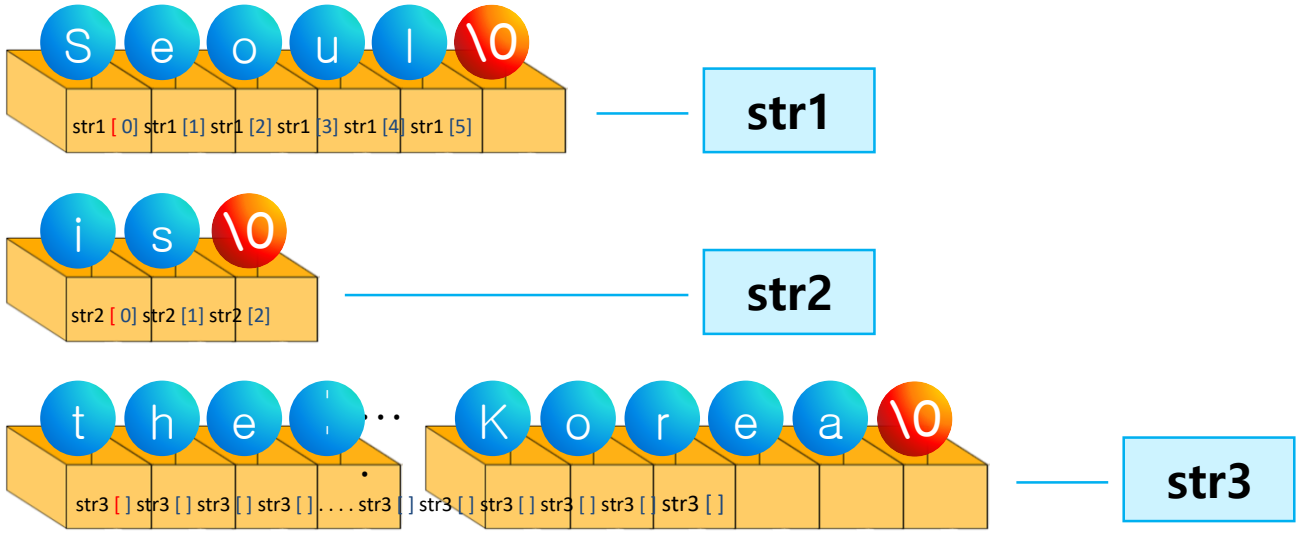


Output of Strings

```
char str [] = “abc”;
printf (“%s”, str );
```

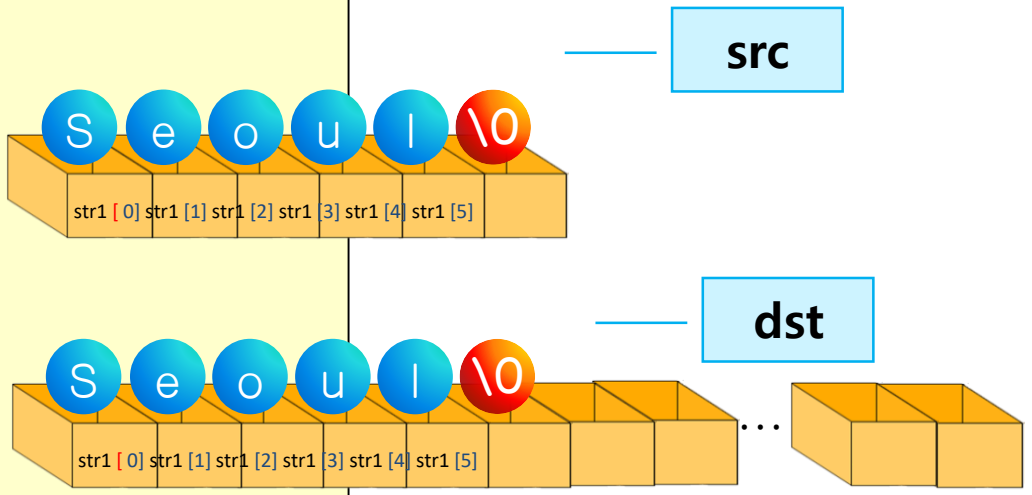



```
#include < stdio.h >

int main( void )
{
    char str1[6] = "Seoul" ;
    char str2[3] = { ' i ' , 's' , '\0' };
    char str3[] = "the capital city of Korea." ;
    printf ( "%s %s %s\n" , str1, str2, str3);
    return 0;
}
```



String Copy

```
#include < stdio.h >

int main( void )
{
        char src [] = “Seoul" ;
        char dst [100];

        int i ;
        printf ( " Original string =%s\n" , src );
        for (i = 0; src[i] != '\0' ; i++)
            dst [ i ] = src [ i ];

        dst [ i ] = '\0' ;
        printf ( " Copied string =%s\n" , dst );

        return 0;
}
```




```
Original string = Seoul
Copied string = Seoul
```

Counting String Length

```c
// Program to find the length of a string
#include < stdio.h >

int main( void )
{
        char str [30] = "C language is easy";
        int i = 0;

        while ( str [ i ] != 0) {
             i ++;
        }
        printf ( "The length of the string \"%s\" is %d.\n" , str , i );
        return 0;
}
```

Changing a Character Array

- Correct Way

1. Assign the desired character to each array element individually
   - a sure but very inconvenient way.
2. Copy a **string** into a **character array** using the library function **strcpy**().

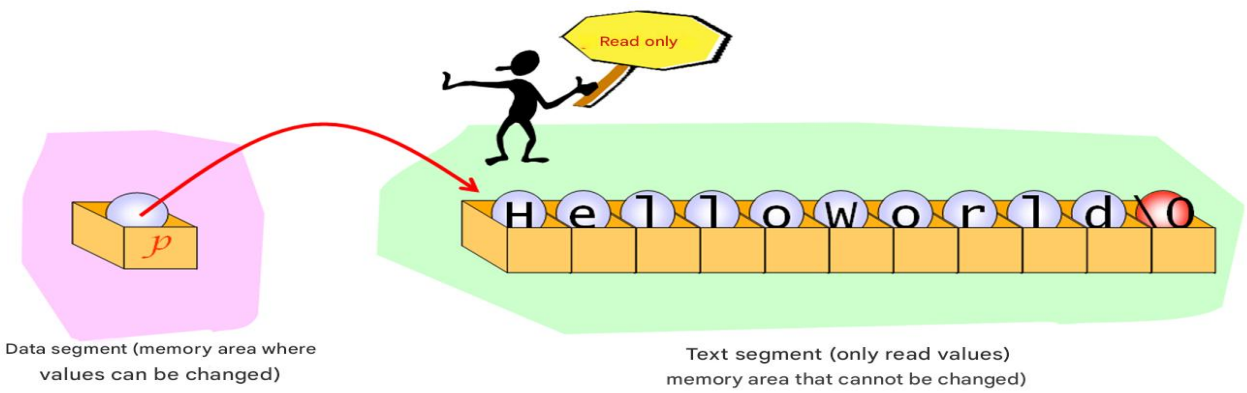
- Wrong Way

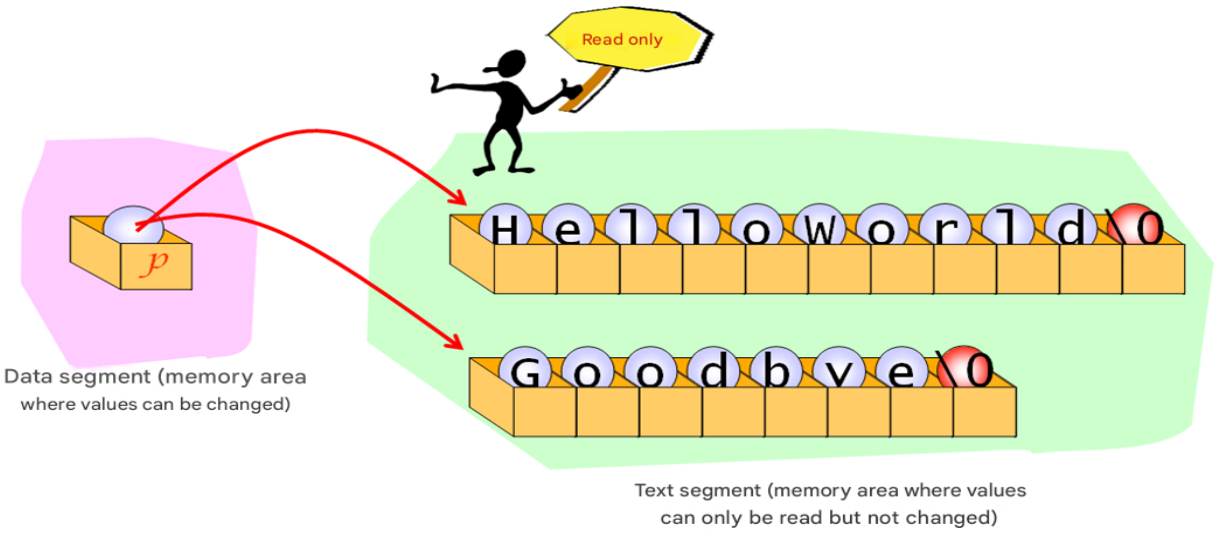
1. char str[10] = "Hello";
   str = "World";
   - The name of an array is a pointer constant that points to the array, it cannot be changed.
2.

```
char *p = "HelloWorld";      // Wrong (Read Only)
char p[20] = "HelloWorld";  // Correct
strcpy (p, “Goodbye”);
```

String Constant

- **String** constant: a **string** included in the program source code, such as "HelloWorld". **String** constants are stored in the text segment.
- char *p = "HelloWorld"; - p points to a **string** constant.
- **strcpy**(p, "Goodbye"); causes an error, because the text segment cannot be changed.
- p = "Goodbye"; (reassigning p itself) is OK - p can point to a different **string**.



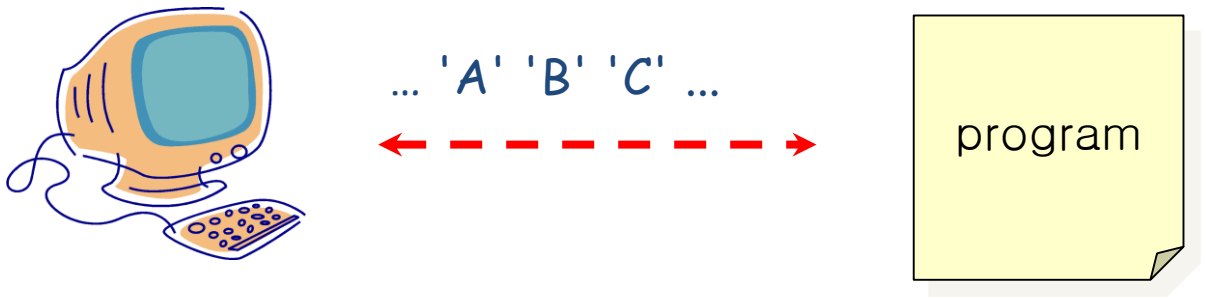


How to read & write characters

Character input/output library

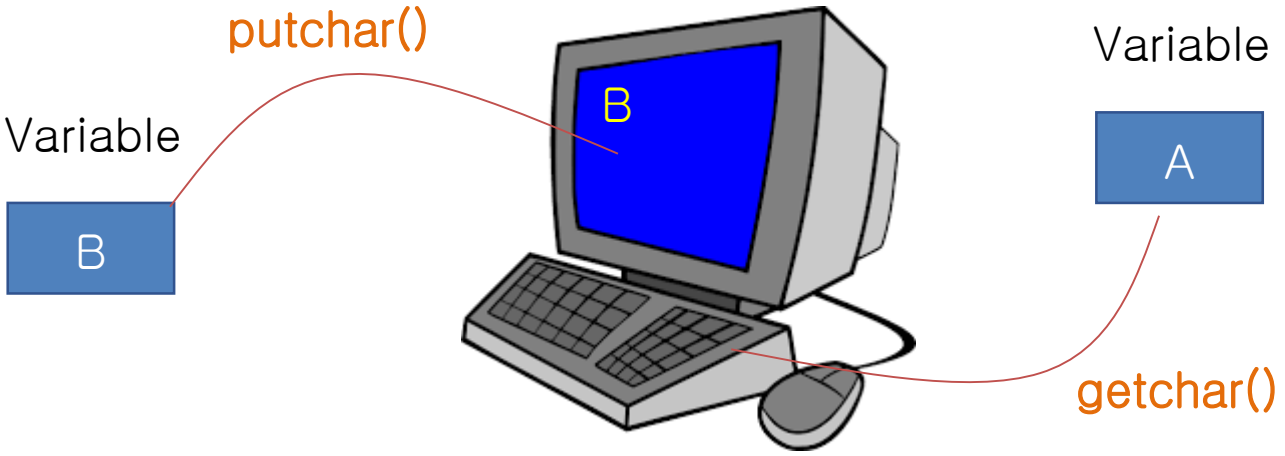
Character Input/Output Library

Input/output functions	explanation
int getchar(void)	Reads and returns a single character.
void putchar(int c)	Prints the character stored in variable c.
int getch(void)	Reads and returns a single character (without using a buffer).
void putch(int c)	Prints the character stored in variable c (without using a buffer).



Character Input/Output: getchar(), putchar()

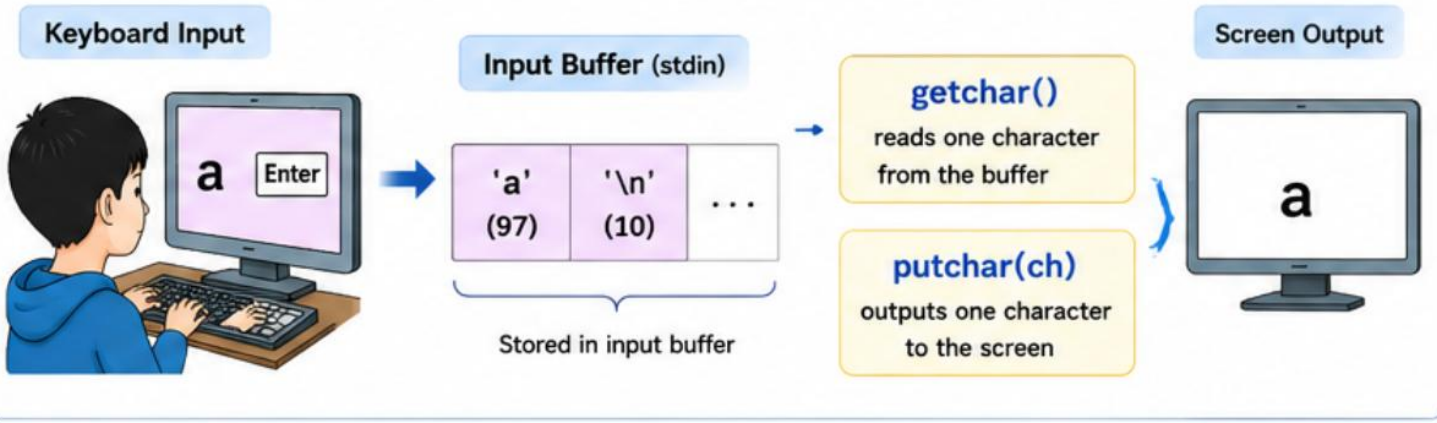
- EOF (End Of File): a special symbol indicating the end of input. EOF is the integer -1.
- getchar() returns EOF if the end of input is reached, or an input error occurs.
- The return type of getchar() is "int" (not char), to distinguish a real ASCII code from EOF.



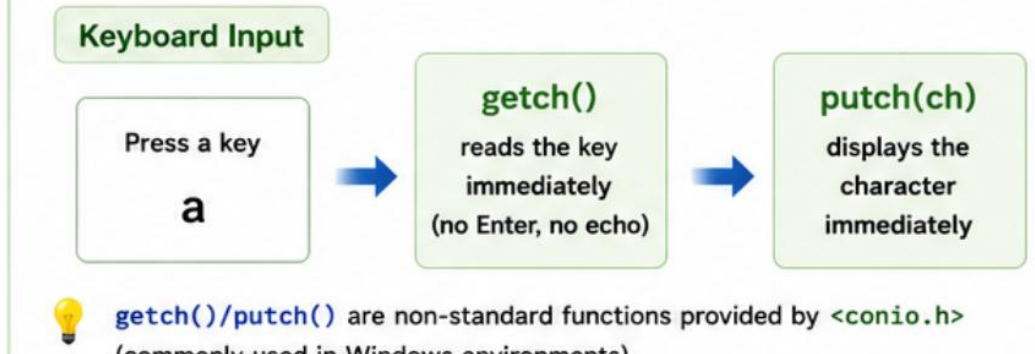
Character Input/Output: getchar(), putchar()

Function	Prototype	Description	Header / Notes
getchar()	`int getchar(void);`	• Reads one character from the standard input (keyboard). • Returns the character as an `int`, • Enter key generates `'\n'` (newline).	`#include <stdio.h>` Standard C function.
putchar(ch)	`int putchar(int ch);`	• Writes one character (`ch`) to the standard output (screen). • Returns the character written as an `int`, or EOF on error.	`#include <stdio.h>` Standard C function.
getch()	`int getch(void);`	• Reads one character from the keyboard without waiting for Enter. • The character is not echoed (not displayed).	`#include <conio.h>` Non-standard (Windows).
putch(ch)	`int putch(int ch);`	• Writes one character to the screen without buffering. • Returns the character written, or EOF on error.	`#include <conio.h>` Non-standard (Windows).

How getchar() and putchar() work



How getch() and putch() work (Windows)

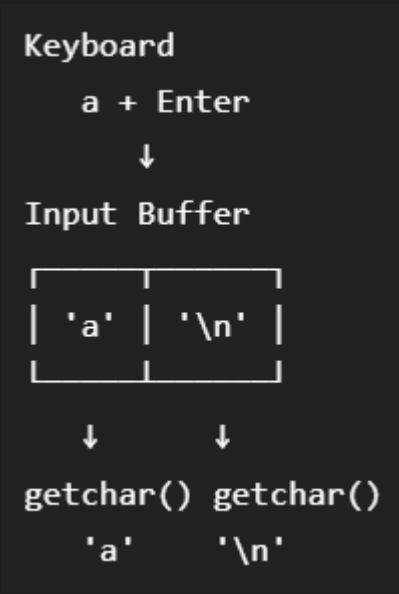


getch()/putch() are non-standard functions provided by <conio.h> (commonly used in Windows environments).

getchar() / putchar() Example

```c
#include < stdio.h >
int main( void )
{
    int ch ;  // Note the use of int, not char
    while ( ( ch = getchar ()) != EOF )
        putchar ( ch );
    return 0;
}


// To terminate while loop
    while ( ( ch = getchar ()) != EOF && ch != ‘q’)
```

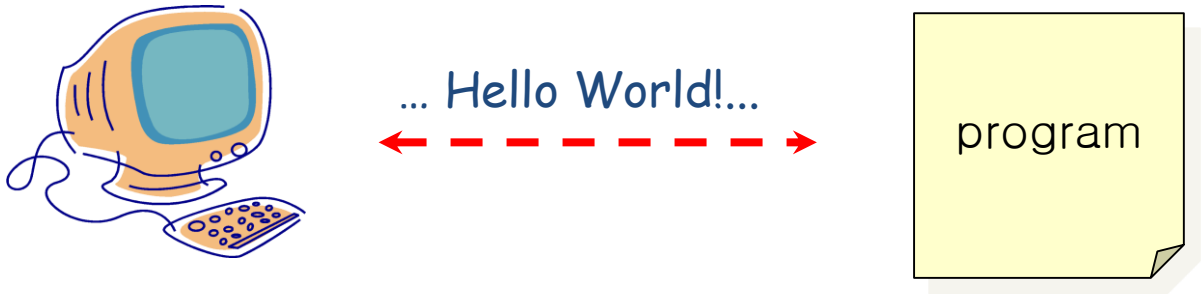
getchar() : returns EOF if the end of input is reached, or an input error occurs




```
a
a
b
b
^Z
```

String Input/Output Library

Input/output functions	explanation
char * gets_s(char *s, int size)	Read a line of string and store it in the character array s[].
int puts(const char *s)	Prints a single line of string stored in array s[].
scanf("%c", &c)	Read one character and store it in variable c.
printf("%c", c);	Prints the character stored in variable c.



String Input/Output Library: gets_s(), puts()

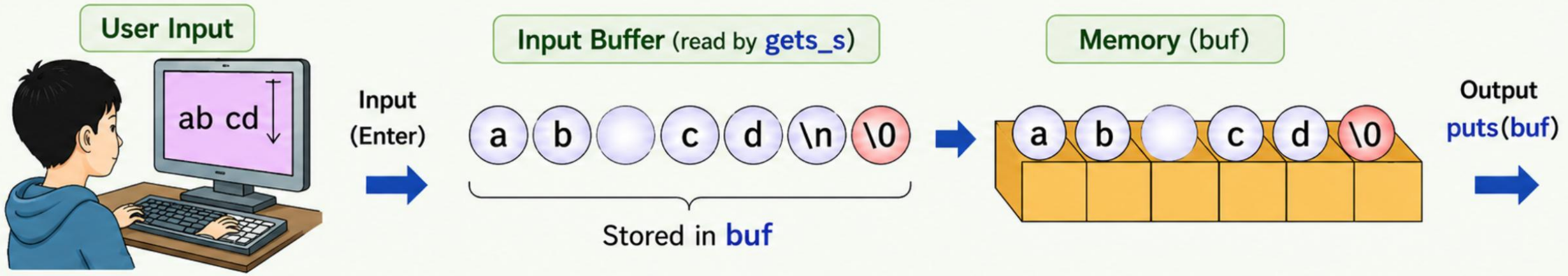
```
char buf[100];

gets_s(buf, 100);      // reads a line of input

puts(buf);             // prints the line
```

gets_s(buf, size)

- Reads a line (until Enter) from the user.
- Stores it in **buf** (including '\0').

puts(buf)

- Prints the string stored in **buf**.
- Automatically adds a newline at the end.



Summary

- gets_s(buf, size): Reads a line from input and stores it in **buf** (appends '\0').
- puts(buf): Prints the string in **buf** and adds a newline.

String Input/Output Library: printf(), scanf()

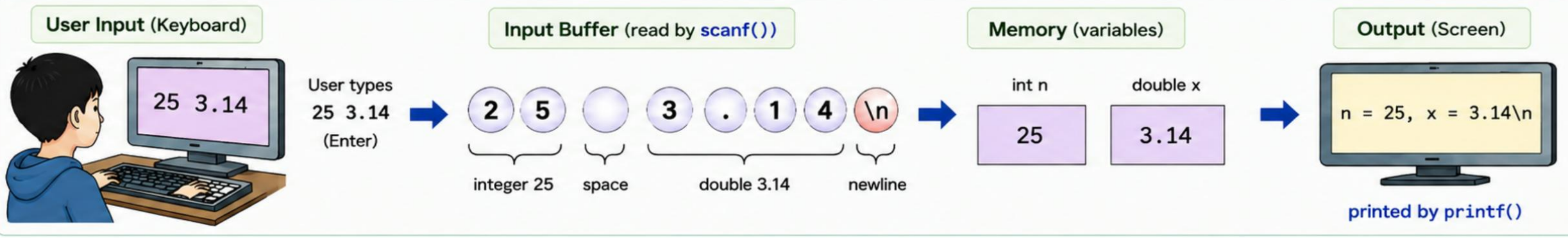
```
printf("format string", arg1, arg2, ...);

scanf("format string", &var1, &var2, ...);
```

printf()

- Prints formatted output to the screen.
- Returns the number of characters printed.

scanf()

- Reads formatted input from the keyboard.
- Stores values in the provided variables.
- Returns the number of items successfully read.



printf() Example

```
int n = 25;
double x = 3.14;
printf("n = %d, x = %.2f\n", n, x);
```

Common Format Specifiers

%d	→	integer
%f	→	floating point
%c	→	character
%s	→	string
%lf	→	double

scanf() Example

```
int n;
double x;
scanf("%d %lf", &n, &x);
```

Common Format Specifiers

%d	→	integer
%f	→	float
%lf	→	double
%c	→	character
%s	→	string

Summary

- `printf()` : Prints formatted output to the screen using a format string.
- `scanf()` : Reads formatted input from the keyboard and stores it in variables (use & before variable names).

Tip

In `scanf()`, spaces (or \t, \n) in the format string skip any amount of whitespace (space, tab, newline).

String Input/Output: scanf("%s", ...)

```c
#include < stdio.h >

int main( void )
{
    char name[100];
    char address[100];

    printf ( "Enter your name: " );
    scanf ( "%s" , name);
    printf ( "Enter your current address: " );
    scanf ( "%s" , address);

    printf ( "Hello, Mr. %s who lives in %s.\n" , name, address);
    return 0;
}
// Caution: scanf("%s", ...) does not save the FULL address
// if it contains spaces - it stops at the first space!
```

Character Processing Library: <ctype.h>

- Check or convert characters.

Function	explanation
isalpha (c)	Is c an English letter ? (az, AZ)
isupper (c)	Is c uppercase ? (AZ)
islower(c)	c lowercase ? (az)
isdigit (c)	Is c a number ? (0-9)
isalnum(c)	c a letter or a number ? (az, AZ, 0-9)
isxdigit(c)	Is c a hexadecimal number ? (0-9, AF, af)
isspace(c)	Is c a space character ? (' ', '\n', '\t', '\v', '\r')
ispunct(c)	Is c a punctuation character ?
isprint(c)	C a printable character ?
iscntrl(c)	Is c a control character ?
isascii(c)	c an ascii code ?

Function	explanation
toupper(c)	c to uppercase .
tolower(c)	c to lowercase .
toascii(c)	c to ASCII code .

Example: Convert Lowercase to Uppercase

```c
#include < stdio.h >
#include < ctype.h >

int main( void )
{
    int c;
    while ((c = getchar ()) != EOF)
    {
        if ( islower (c) )
            c = toupper (c);
        putchar (c);
    }
    return 0;
}
```

How to read & write string

String input/output library

String Processing Library: <string.h>

- **String** functions are declared in **string.h**
  - you must include **string.h** to use them.

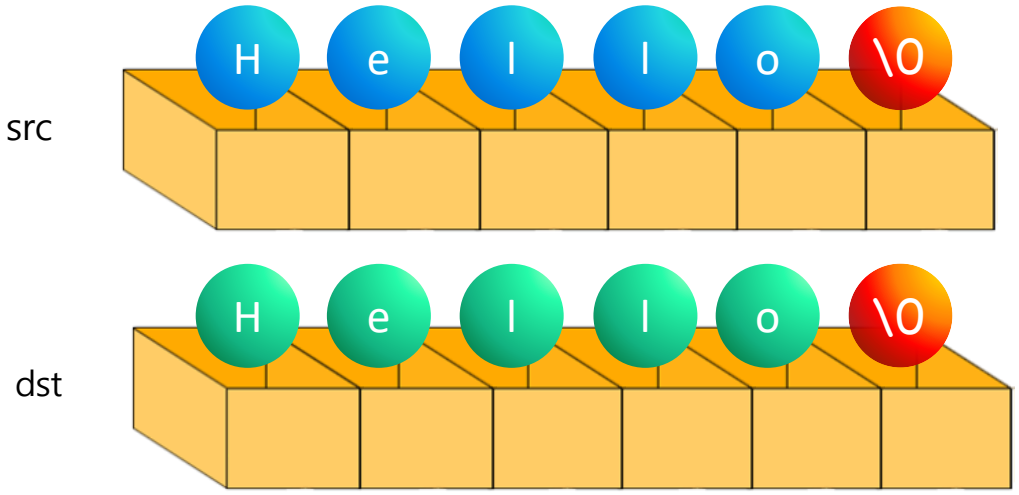
Function	explanation
strlen(s)	Finds the length of string s
strcpy(s1, s2)	Copy s2 to s1
strcat(s1, s2)	Paste s2 to the end of s1
strcmp(s1, s2)	Compare s1 and s2
strncpy(s1, s2, n)	Copy at most n characters from s2 to s1
strncat(s1, s2, n)	Paste at most n characters from s2 to the end of s1
strncmp(s1, s2, n)	Compares s1 and s2 up to n characters
strchr(s, c)	Finds the character c in string s
strstr(s1, s2)	Find string s2 in string s1

String Processing Library: String Copy

Syntax strcpy()

yes

```
char dst[6];
char src[6]="Hello";
strcpy(dst, src);                    // Copy src to dst.
```

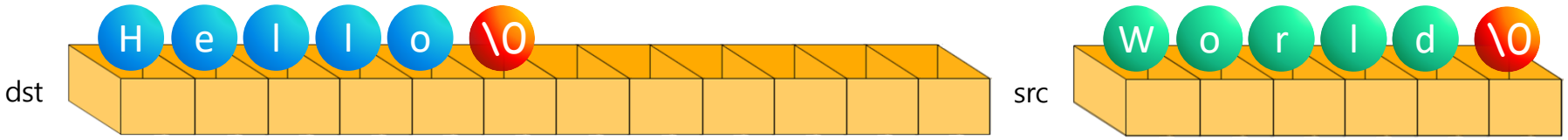


String Processing Library: String Concatenation

Syntax strcat()

yes

```
char dst[12]= "Hello";
char src[6] = "World";
strcat(dst, src);              // dst becomes "HelloWorld".
```



strcpy() and strcat()

```c
#include < string.h >
#include < stdio.h >

int main( void )
{
    char string[80];
    strcpy ( string, "Hello world from " );
    strcat (string, " strcpy " );
    strcat ( string, "and " );
    strcat ( string, " strcat!" );
    printf ( "string = %s\n" , string );
    return 0;
}
```

String Processing Library: String Comparison (strcmp())

Syntax `strcmp( )`

```
int result = strcmp("dog", "dog");
```

strcmp() compares strings s1 and s2 lexicographically.

- Comparison starts at the first character.
- If all compared characters are equal and both strings end, the strings are equal.
- If a difference is found, the comparison stops and the result is returned.

Return Value	Relationship between s1 and s2	Meaning
< 0	s1 comes before s2. (s1 is lexicographically smaller than s2)	The first differing character in s1 has a smaller ASCII value than in s2.
0	s1 == s2 (strings are equal)	All characters are the same and both strings end.
> 0	s1 comes after s2. (s1 is lexicographically larger than s2)	The first differing character in s1 has a larger ASCII value than in s2.

If the strings are equal, strcmp( ) returns 0.

Remember

strcmp() does NOT change the original strings.
It only compares them.

Notes / Example

- Strings are compared based on ASCII values of characters.
- strcmp() is similar but compares up to n characters.

```
strcmp("apple",  "app")    > 0    // 'l' (108)> '\0' (0)
strcmp("cat",    "dog")    < 0    // 'c' (99) <  'd' (100)
strcmp("same",   "same")   == 0   // equal
```

How comparison works (example)

s1:	d	o	g	\0
s2:	d	o	g	\0

All characters match and both end → result is 0.

strcmp(): Comparing Strings

```
int result = strcmp (s1, s2);

if ( result < 0 )
    printf ( "%s is less than %s\n" , s1, s2);
else if ( result == 0 )
    printf ( "%s and %s are the same.\n" , s1, s2);
else
    printf ( "%s is greater than %s\n" , s1, s2);
// strcmp() returns 0 when the strings are equal.
```

String Processing Library: Search a Character (strchr())

```
char *p = strchr("dog", 'g');
```

Return the address of character 'g'

What it does

- Searches for the first occurrence of character **ch** in the string **str**.
- Returns a pointer to that character in the string.
- If not found, returns **NULL**.

Return Value	Meaning	Explanation
Pointer to character	Address of the first occurrence of ch in str.	A pointer to the character found in the string.
NULL	ch is not found in str.	No such character exists in the string.

Example

```
char *p = strchr("dog", 'g');
if (p != NULL)
   printf("Found at index %ld\n", p - "dog");
else
   printf("Not found\n");
```

String index

0	1	2
d	o	g

p points here

More Examples

strchr("banana", 'a')

0	1	2	3	4	5
b	a	n	a	n	a

Returns address of index 1

strchr("banana", 'n')

0	1	2	3	4	5
b	a	n	a	n	a

Returns address of index 2

strchr("banana", 'z')

0	1	2	3	4	5
b	a	n	a	n	a

Returns NULL

Tips

- strchr() is case-sensitive.
- Include <string,h> to use strchr().
- The returned pointer points inside the original string.

String Processing Library: Search a string (strstr())

Syntax — strstr()

```
char *p = strstr("dog and cat", "cat");
```

"dog and cat" → string s

"cat" → substring sub

p points to the first occurrence of "cat" in s

What it does

- Searches for the first occurrence of substring **sub** in string **s**.
- If found, returns the address where **sub** begins.
- If not found, returns **NULL**.

Return Value	Meaning
Pointer to char	Address of the first character of the first occurrence of sub in s.
NULL	sub is not found in s.

Example

```
char *p = strstr("dog and cat", "cat");
```

Index:	0	1	2	3	4	5	6	7	8	9	10
String s:	d	o	g		a	n	d		c	a	t	\0

p points here (address of index 8)

More Examples

`strstr("hello world", "world")`

Index:	0	1	2	3	4	5	6	7	8	9	10
String s:	h	e	l	l	o		w	o	r	l	d	\0

Returns address of index 6

`strstr("hello world", "lo w")`

Index:	0	1	2	3	4	5	6	7	8	9	10
String s:	h	e	l	l	o	w	w	o	r	l	d	\0

Returns address of index 3

`strstr("hello world", "abc")`

Index:	0	1	2	3	4	5	6	7	8	9	10
String s:	h	e	l	l	o		w	o	r	l	d	\0

Returns NULL

Tips

- strstr() is case-sensitive.
- Include <string.h> to use strstr().
- The returned pointer points inside the original string s.

strchr(): Text Search

```c
#include < string.h >
#include < stdio.h >

int main( void )
{
    char s[] = "language";
    char c = 'g';
    char *p;
    int loc;

    p = strchr (s, c);
    if ( p == NULL )
        printf ( "%c not found\n" , c );
    else {
        loc = ( int )(p - s);
        printf ( "First %c in %s found at %d\n" , c, s, loc);
    }
    return 0;
}
```

To check whether a string contains a specific character, use strchr(). To search for a whole substring, use strstr().

String Processing Library: String Tokenization (strtok())

Syntax

```
char *p = strtok("Hello World!", " ");
```

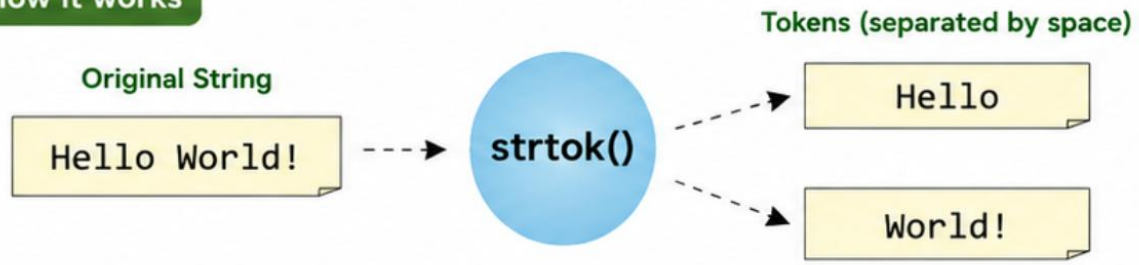
function — string to be tokenized — separator (delimiters)

Splits a string into words using spaces.

What it does

- Splits a string into tokens (words) using the specified separator characters.
- On the first call, pass the string to be tokenized.
- On subsequent calls, pass **NULL** as the first argument to get the next tokens.
- Returns a pointer to the next token found.
- When no more tokens remain, returns **NULL**.

How it works



Example

```
char *p = strtok("Hello World!", " ");

while (p != NULL) {
   printf("%s\n", p);      // print token
   p = strtok(NULL, " ");  // get next token
}
```

Output

```
Hello
World!
```

Examples with different separators

String (s)	Separator	Tokens returned (in order)
"one,two,three"	","	one → two → three
"a;b;c"	";"	a → b → c
"red|green|blue"	"|"	red → green → blue

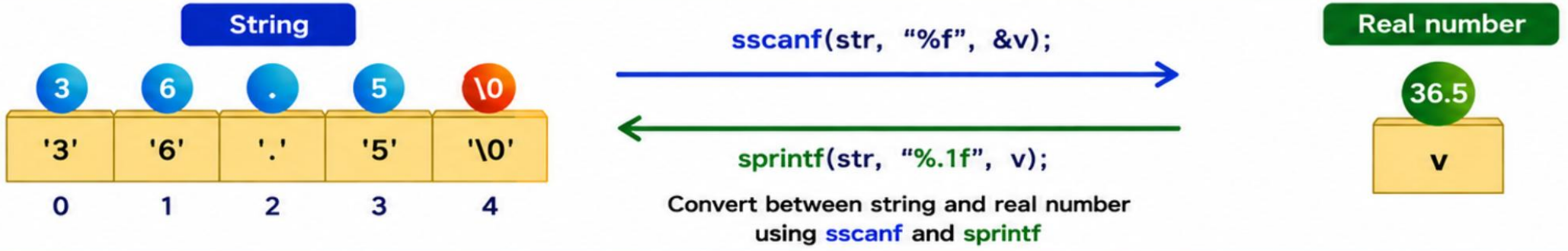
Tips

- Include <string.h> to use strtok().
- Consecutive separators are treated as one.
- The original string is modified: separators are replaced with '\0'.
- Not thread-safe (uses internal static data).

strtok(): String Seperation

```
char[] s = “Man is immortal, because he has a soul” ;
t1 = strtok (s, “ ” );        // First token :      Man            “ “ is separator.
t2 = strtok (NULL, “ ” ); // Second token : is
t3 = strtok (NULL, “ ” ); // Third token :     immortal,
t4 = strtok (NULL, “ ” ); // Fourth token :   because
t5 = strtok (NULL, “ ” ); // Fifth token :       he
t6 = strtok (NULL, “ ” ); // Sixth token :      has
t7 = strtok (NULL, “ ” ); // Seventh token : a
t8 = strtok (NULL, “ ” ); // Eighth token :    soul
t9 = strtok (NULL, “ ” ); // Ninth token :      NULL
```

String Conversion: String to Number, and vice versa



sscanf() : Convert a string to a number

- Reads the number in str according to the format and stores it in the variable pointed by &v. (sscanf is string input)

str	→	Original string containing the number
"%f"	→	Format specifier to read float (real number)
&v	→	Address of variable v to store the read value

sprintf() : Convert a number to a string

- Creates a string in str according to the format using the value of v. (sprintf is string output)

str	→	Character array to store the (numeric) string
"%.1f"	→	Format specifier for float (e.g., %.1f, %f) (specific output format)
v	→	Real number value to be converted into a string

Functions to convert a string to a number (in <stdio.h>)

Function	Explanation
int atoi(const char *str);	Convert string to int type.
double atof(const char *str);	Convert string to double type.

Example : Using atoi and atof

```c
#include <stdio.h>
#include <stdlib.h>

int main(void) {
   char s1[100] = "123";
   char s2[100] = "45.67";

   int i = atoi(s1);      // convert string to int
   double d = atof(s2);   // convert string to double

   printf("i = %d\n", i);
   printf("d = %.2f\n", d);
   return 0;
}
```

Output

```
i = 123
d = 45.67
```

Common Mistakes

- A C string is just a char array ending in the **NULL terminator** '\0'
  - forgetting it causes garbage output.
- **strcpy**/**strcat** never check destination size
  - writing past the buffer is a classic **buffer overflow** bug.
- Comparing strings with **==** compares addresses, not contents
  - always use **strcmp()** instead.
- A string literal like "Hello" cannot be modified
  - use **char arr[] = "Hello"** to get a modifiable copy.

Q & A

Lab & Quiz: right after this - see you in the practice session!